



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Experiment No.6
To perform Network packet forensics using Network Miner
Date of Performance:
Date of Submission:



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Aim: To perform network traffic forensics using network miner

Objective: To extract the artifact from the network traffic using network miner tool

Theory:

Packet Capture or PCAP (also known as libpcap) is an application programming interface (API) that captures live network packet data from OSI model Layers 2-7. Network analyzers like Wireshark create .pcap files to collect and record packet data from a network. PCAP comes in a range of formats including Libpcap, WinPcap, and PCAPng.

These PCAP files can be used to view TCP/IP and UDP network packets. If you want to record network traffic then you need to create a .pcapfile. You can create a .pcapfile by using a network analyzer or packet sniffing tool like Wireshark or tcpdump.

PCAP is a valuable resource for file analysis and to monitor your network traffic. Packet collection tools like Wireshark allow you to collect network traffic and translate it into a format that's human-readable. There are many reasons why PCAP is used to monitor networks. Some of the most common include monitoring bandwidth usage, identifying rogue DHCP servers, detecting malware, DNS resolution, and incident response.

For network administrators and security researchers, packet file analysis is a good way to detect network intrusions and other suspicious activity. For example, if a source is sending the network lots of malicious traffic, you can identify that on the software agent and then take action to remediate the attack.

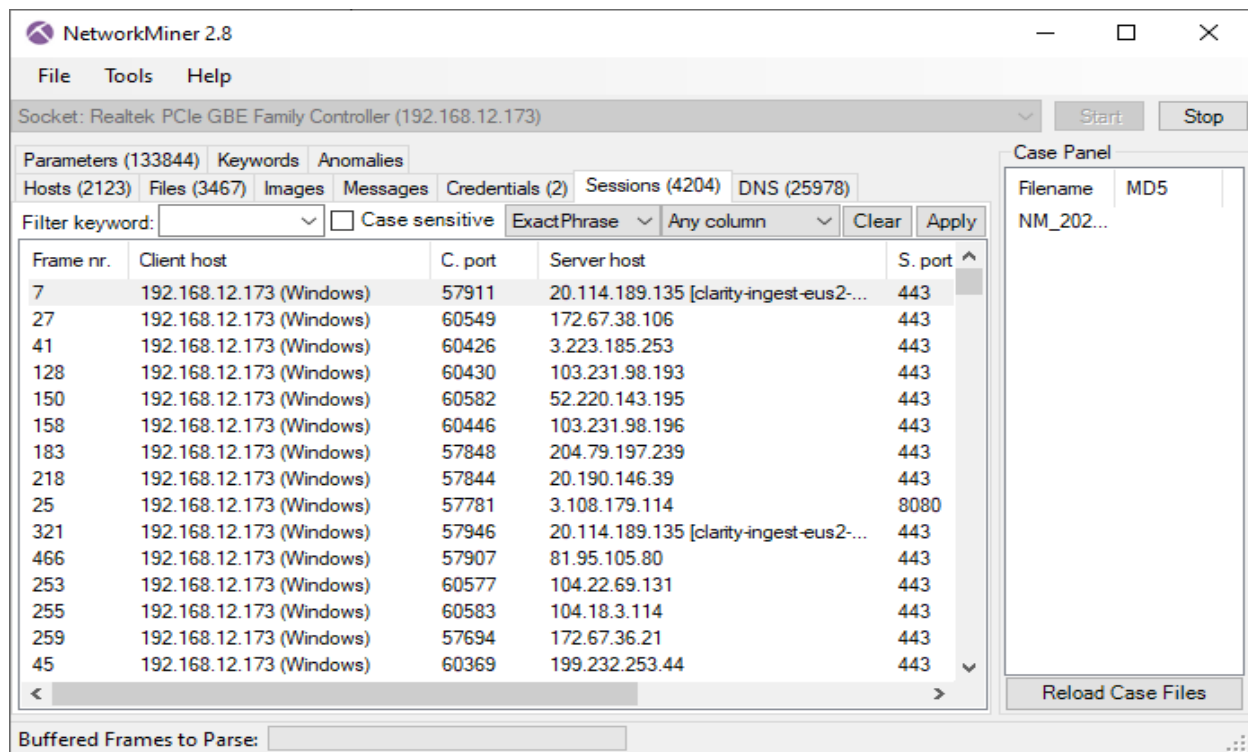
Network Miner

Network Miner is an open source network forensics tool that extracts artifacts, such as files, images, emails and passwords, from captured network traffic in PCAP files. Network Miner can also be used to capture live network traffic by sniffing a network interface. Detailed information about each IP address in the analyzed network traffic is aggregated to a network host inventory, which can be used for passive asset discovery as well as to get an overview of which devices that are communicating. Network Miner is primarily designed to run in Windows, but can also be used in Linux.

Network Miner has, since the first release in 2007, become a popular tool among incident response teams as well as law enforcement. Network Miner is today used by companies and organizations all over the world. Figure below shows the screenshot of the network miner tool.



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Network Miner can be used as a passive network sniffer/packet capturing tool in order to detect operating systems, sessions, hostnames, open ports etc. without putting any traffic on the network.

Network Miner can also parse PCAP files for off-line analysis and to regenerate/reassemble transmitted files and certificates from PCAP files.

Process:

- Step 1. Download the network Miner tool from the website [NetworkMiner - The NSM and Network Forensics Analysis Tool](https://www.netresec.com/) (netresec.com)
- Step 2. Install the tool onto your system
- Step 3. Connect your system to the internet connection
- Step 4. Open the Network Miner tool
- Step 5. Extract the Network traffic artefact using Network Miner tool
- Step 6. Create a report based on the artefact extracted



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Output:

The screenshot displays the NetworkMiner 2.9.0 application window. The main pane shows a list of network frames with columns: Frame nr., Client host, C. port, Server host, S. port, Protocol (application layer), and Start time. The data is filtered by the keyword 'rbf'. The right sidebar shows the 'Case Panel' with a list of files and folders. The bottom pane shows a detailed view of the selected frame, including DNS query and response details.

Frame nr.	Client host	C. port	Server host	S. port	Protocol (application layer)	Start time	IRTT (ms)
77	10.251.23.139	35383	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
103	10.251.23.139	35384	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
109	10.251.23.139	35386	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
110	10.251.23.139	35385	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
125	10.251.23.139	35388	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
126	10.251.23.139	35387	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
133	10.251.23.139	35389	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	
137	10.251.23.139	35390	86.66.0.227 [general.rbfidsl.neufbox.neuf.fr]	80	HTTP	1970-01-01 00:01:56 UTC	

Conclusion: Using the **NetworkMiner** tool to extract artifacts from network traffic allows investigators to capture crucial information such as files, images, and credentials transmitted over the network. This aids in identifying malicious activity, unauthorized data transfers, or other security breaches. Analyzing network traffic artifacts provides vital evidence to support a digital forensics investigation.